

Operating System Simulation Project Report

1. Introduction

The project **EduOS-25311351009** was developed to simulate fundamental Operating System (OS) concepts using **C** and **Python**. It demonstrates process management, thread handling, inter-process communication (IPC), scheduling, and deployment workflows. GitHub was used for version control and project collaboration.

2. Project Structure

c_core Folder

- Contains the main C source code modules:
 - Process management
 - Thread management
 - IPC communication
 - Simulation coordination

controller Folder

- Python scripts for controlling the simulation (start/stop operations).

python_scheduler Folder

- Implements **Round Robin scheduling** and **Gantt chart visualization**.
- **Other Files**
 - `.gitignore`: Excludes temporary/compiled files from GitHub.
 - `README.md`: Documentation with objectives, setup, and execution steps.
 - `create_structure.py`: Script to auto-generate project folder structure.
 - `pcb_snapshot.json`: Stores simulated Process Control Block (PCB) snapshots.

3. Development Workflow

GitHub Commits

- Progressive commits show implementation of:
 - Scheduler logic

- IPC module creation
- Deployment configuration

Deployment Configuration

- Makefile and GitHub deployment setup for execution workflow.

4. Key Modules & Screenshots

IPC Module

Demonstrates sending and receiving messages between processes.

Example:

```
c
void send_message(char message[]) {
    printf("Sending message: %s\n", message);
}
void receive_message(char message[]) {
    printf("Received message: %s\n", message);
}
```

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Thread Manager

Simulates thread handling using C structs.

Example:

```
c
typedef struct {
    int tid;
    char status[20];
} Thread;
```

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Execution Screenshot

- Shows compilation attempts and terminal output during simulation runs.

5. Achievements

-  Implemented **process and thread management** in C
-  Built **IPC communication** functions
-  Designed **Round Robin scheduler** with visualization in Python
-  Used **GitHub version control** for iterative development
-  Configured **deployment workflow** with Makefile

6. Conclusion

The project successfully demonstrated **basic OS concepts** through simulation. By combining **C for low-level modules** and **Python for scheduling and visualization**, the assignment achieved its objectives. The integration of GitHub commits, deployment configuration, and structured folder organization reflects good software engineering practices.